

Joint AC Validation of the LinDistFlow–Softplus

Formulation: Case 2-1

Supplement to *Source-Attributed DLMP Settlement with Price-Anticipating Load Response*

1 Validation objective and models

This supplement examines the combined effect of the LinDistFlow approximation and Softplus regularization by comparing the resulting nodal prices directly with full nonlinear AC optimal power flow (OPF). The test uses Case 2-1 of Study I, covering all 24 hours and all 32 non-root buses of the 33-bus radial feeder. An independent AC power-flow calculation also checks the physical feasibility of Model C's dispatch.

The three models share the hourly physical demand, flexible-load utility functions and decision bounds, fixed power factors, root voltage, and network limits. Bus 1 is the root; the power and voltage bases are 100 MVA and 13.3 kV. The non-root bus and branch sets are $\mathcal{N}^+$ and $\mathcal{E}$. Let p_i denote active demand and $q_i = \beta_i p_i$, where β_i is the fixed reactive-to-active demand ratio. Fixed demands retain the input data, and flexible demands lie in the same admissible set $\mathcal{Y}$ as in Study I. Their utility is $U_i(p_i) = \alpha_i p_i - \frac{1}{2} \beta_{u,i} p_i^2$. The root squared voltage is $v_1 = 1$; the voltage-magnitude lower limit is 0.93 p.u., and the sending-end active-flow limits are those of Case 2-1. Model C retains $(\kappa_C, \kappa_V, \varepsilon) = (10^8, 10^6, 10^{-6})$.

1.1 Hard and regularized LinDistFlow models

Model B is the hard-constrained LinDistFlow OPF. With the reduced network maps $\mathbf{v} = v_1 \mathbf{1} - A^V \mathbf{p}$ and $\mathbf{f} = A^C \mathbf{p}$, its objective and network constraints are

$$J_{\text{LF}}(\mathbf{p}) = \pi^E \mathbf{1}^\top \mathbf{p} + \frac{\pi^E}{v_1} \mathbf{p}^\top H^L \mathbf{p} - \sum_{i \in \mathcal{F}} U_i(p_i), \quad (1)$$

$$\mathbf{v} \geq \underline{\mathbf{v}}, \quad \mathbf{f} \leq \bar{\mathbf{f}}, \quad \mathbf{p} \in \mathcal{Y}. \quad (2)$$

Here π^E is the wholesale energy price, H^L is the quadratic loss matrix, $\mathcal{F}$ is the flexible-bus set, and $\underline{\mathbf{v}}$ contains squared voltage limits. Model C replaces the two network inequalities by smooth objective penalties while retaining $\mathbf{p} \in \mathcal{Y}$:

$$J_C(\mathbf{p}) = J_{\text{LF}}(\mathbf{p}) + \kappa_V \sum_{i \in \mathcal{N}^+} \varphi_\varepsilon(\underline{v}_i - v_i) + \kappa_C \sum_{e \in \mathcal{E}} \varphi_\varepsilon(f_e - \bar{f}_e), \quad \varphi'_\varepsilon(z) = s_\varepsilon(z) = \varepsilon \log(1 + e^{z/\varepsilon}). \quad (3)$$

The nodal prices in both models have the common form

$$\boldsymbol{\pi}^{\text{LF}} = \pi^E \mathbf{1} + \frac{2\pi^E}{v_1} H^L \mathbf{p} + (A^V)^\top \boldsymbol{\mu} + (A^C)^\top \boldsymbol{\nu}. \quad (4)$$

For B, $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are the hard-constraint multipliers; for C, they are the penalty derivatives $\mu_i = \kappa_V s_\varepsilon(\underline{v}_i - v_i)$ and $\nu_e = \kappa_C s_\varepsilon(f_e - \bar{f}_e)$. Each price vector is evaluated at its own optimized dispatch.

1.2 Full nonlinear AC benchmark

For a directed branch (i, j) , let P_{ij}, Q_{ij} be sending-end power, $v_i = |V_i|^2$, and $\ell_{ij} = |I_{ij}|^2$. With resistance r_{ij} , reactance x_{ij} , and child set $\mathcal{C}(j)$, the AC benchmark enforces the complete DistFlow equalities:

$$P_{ij} = p_j + \sum_{k \in \mathcal{C}(j)} P_{jk} + r_{ij} \ell_{ij}, \quad Q_{ij} = \beta_j p_j + \sum_{k \in \mathcal{C}(j)} Q_{jk} + x_{ij} \ell_{ij}, \quad (5)$$

$$v_j = v_i - 2(r_{ij} P_{ij} + x_{ij} Q_{ij}) + (r_{ij}^2 + x_{ij}^2) \ell_{ij}, \quad v_i \ell_{ij} = P_{ij}^2 + Q_{ij}^2. \quad (6)$$

Thus, active and reactive losses and the quadratic voltage correction are retained. The objective is

$$\min J_{\text{AC}} = \pi^E \sum_{j \in \mathcal{C}(1)} P_{1j} - \sum_{i \in \mathcal{F}} U_i(p_i), \quad \text{subject to (5)–(6), } v_i \geq \underline{v}_i, P_{ij} \leq \bar{f}_{ij}, \ell_{ij} \geq 0, \mathbf{p} \in \mathcal{Y}. \quad (7)$$

Root imports already include physical losses. Equations (1) and (7) use per-unit powers and costs normalized by the common power base and one-hour interval.

Write the active and reactive balance residuals as demand plus outgoing power plus losses minus incoming power. For the Lagrangian $J_{AC} + \lambda_P^\top h_P + \lambda_Q^\top h_Q + \dots$, the marginal price for an exogenous demand increase along the fixed-power-factor direction is

$$\pi_i^{AC} = \lambda_i^P + \beta_i \lambda_i^Q, \quad i \in \mathcal{N}^+. \quad (8)$$

Solver scaling is removed before assembling these prices, which are reported in \$/MWh. This definition aligns the AC benchmark with the fixed-power-factor prices of B and C at every non-root bus.

2 Joint validation procedure

For each hour t , B, C, and AC OPF are solved with the shared inputs. The comparisons C–B, B–AC, and C–AC respectively quantify the additional regularization effect within LinDistFlow, the hard LinDistFlow discrepancy, and the combined discrepancy against AC. For a comparison X – Y , define

$$e_{XY,t} = \max_{i \in \mathcal{N}^+} |\pi_{i,t}^X - \pi_{i,t}^Y|, \quad \eta_{XY,t} = 100 \frac{e_{XY,t}}{\max\{1 \text{ \$}/\text{MWh}, \max_{i \in \mathcal{N}^+} |\pi_{i,t}^{AC}|\}}. \quad (9)$$

The reported daily maxima are $\max_t e_{XY,t}$ and $\max_t \eta_{XY,t}$; the latter is the maximum of the hourly normalized errors, not a daily average.

Physical comparisons hold C's optimized demands fixed. A separate backward-current/forward-complex-voltage sweep computes the AC state without adjusting demand or relaxing limits. Both LinDistFlow and AC are evaluated at this same dispatch. The hourly minimum voltage margin and maximum active-flow loading are

$$m_t^X = \min_{i \in \mathcal{N}^+} (\sqrt{v_{i,t}^X} - \sqrt{\underline{v}}), \quad L_t^X = 100 \max_{(i,j) \in \mathcal{E}} \frac{P_{ij,t}^X}{\bar{f}_{ij}}, \quad X \in \{\text{LF}, \text{AC}\}. \quad (10)$$

The maximum voltage-prediction error is $\max_{i,t} |\sqrt{v_{i,t}^{\text{LF}}} - \sqrt{v_{i,t}^{\text{AC}}}|$ under that fixed dispatch.

B uses Gurobi, and C retains the original bounded nonlinear least-squares solution procedure with residual refinement. AC OPF uses SciPy's `trust-constr` solver with analytic derivatives and scaled variables, constraints, and objective. Two AC initial states are constructed from B's dispatch and the flexible-load lower bounds. The accepted local solution with the lower objective is retained. Acceptance requires primal residuals at most 10^{-8} p.u. and normalized KKT residuals at most 10^{-6} ; stationarity and complementarity are normalized by $\max\{1, \pi^E\}$ after restoring the original scaling. Independent AC replay also checks the OPF state.

At the hour of maximum absolute C–AC price error, exogenous demand perturbations are applied at buses 2, 4, and 22, with two step sizes per bus. Active and reactive perturbations follow $(\delta, \beta_i \delta)$; utility coefficients, decision bounds, and other inputs remain fixed. Reoptimization for both signs yields left, right, and central objective differences, providing six checks of the multiplier-based prices. The steps are $\delta_i = \max\{10^{-7}, 10^{-3} \hat{p}_i\}$ p.u. and $\delta_i/2$, where $\hat{p}_i$ is planned demand. A central difference passes when its price discrepancy is no greater than $\max\{0.01 \text{ \$}/\text{MWh}, 0.001 |\pi_i^{AC}|\}$.

3 Results and analysis

Table 1 summarizes the daily price and physical-validation results.

Table 1: Daily joint-validation results for Case 2-1.

Metric	Result
Maximum C–B nodal-price difference	1.06×10^{-10} \$/MWh
Maximum C–AC nodal-price difference	1.286 \$/MWh
Maximum hourly normalized C–AC price error	1.672%
Maximum voltage-magnitude prediction error at C’s dispatch	1.125×10^{-3} p.u.
Maximum AC active-flow loading at C’s dispatch	52.359%
Hours satisfying the tested AC voltage and active-flow limits	24/24

3.1 Price agreement with the AC benchmark

The comparisons show close agreement between Model C’s nodal prices and the full AC benchmark. Across all 24 hours and 32 buses, the maximum hourly normalized error is 1.672% (20:00, bus 18). The maximum absolute price difference is 1.286 \$/MWh (18:00, bus 18). These peaks refer to the two distinct statistics in Eq. (9).

Figure 1 shows that the B–AC and C–AC error curves essentially coincide throughout the day. The maximum C–B difference is only 1.06×10^{-10} \$/MWh. These comparisons show that Softplus regularization introduces no appreciable additional price discrepancy within LinDistFlow. The combined C–AC comparison retains the close agreement of B with AC and directly evaluates the two approximations as used together.

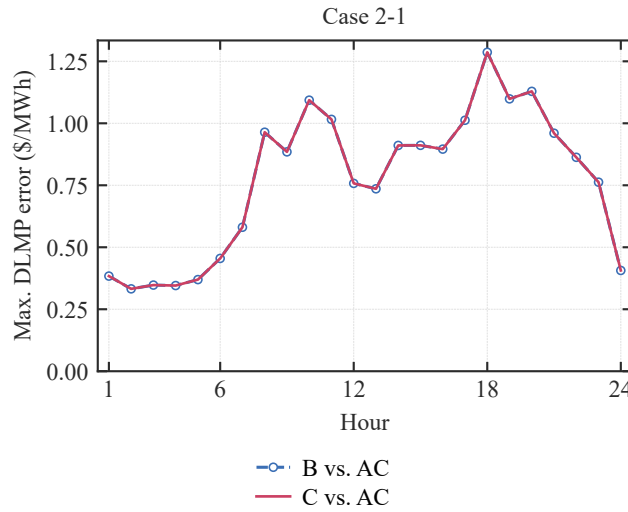


Figure 1: Hourly maximum nodal-price errors against AC OPF in Case 2-1. Hour 24 denotes 00:00.

3.2 AC feasibility and numerical verification

Independent AC replay confirms that C’s dispatch satisfies the tested voltage and sending-end active-flow limits in all 24 hours. Under the same demands, the largest voltage-magnitude prediction error is 1.125×10^{-3} p.u. The AC voltage margin remains positive, and the maximum active-flow loading is 52.359%. Figure 2 shows the daily voltage margins and line loadings relative to their limits. Model C’s dispatch thus remains feasible under the independent nonlinear network calculation.

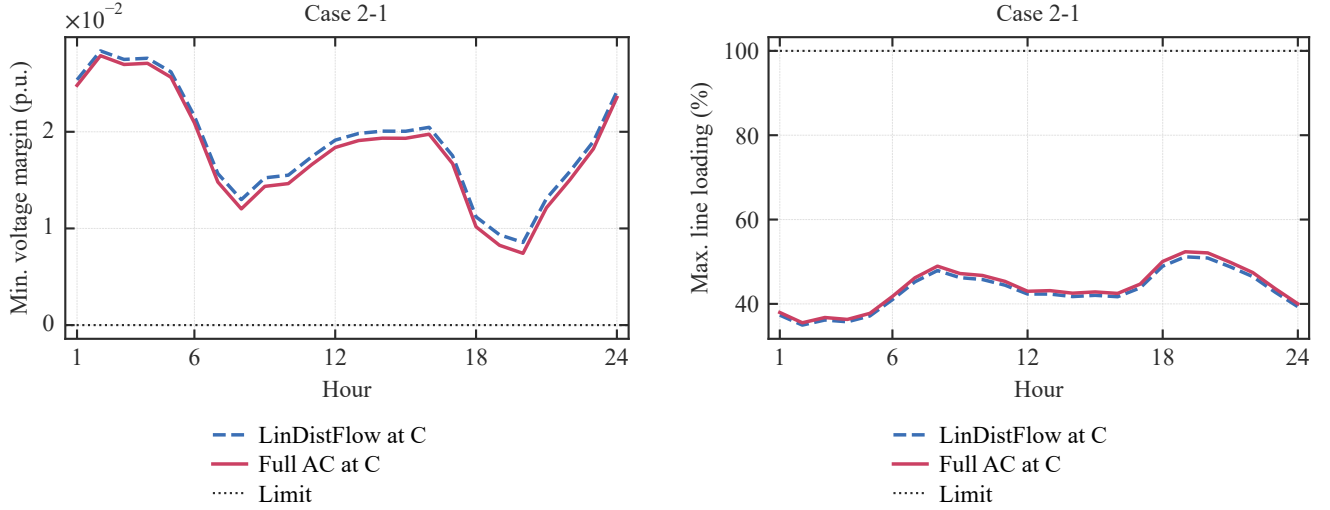


Figure 2: Physical checks at Model C’s fixed dispatch in Case 2-1: minimum voltage-magnitude margin (left) and maximum sending-end active-flow loading (right). The limits are 0 p.u. and 100%, respectively.

All 24 hourly AC benchmarks pass the numerical acceptance criteria, with both starts accepted in each hour. The largest primal residual is 1.24×10^{-16} p.u.; the largest normalized stationarity and complementarity residuals are 1.29×10^{-9} and 3.58×10^{-10} . Independent replay of AC OPF’s dispatch agrees with its branch powers, squared voltages, and squared currents to within 7.93×10^{-14} p.u. The AC replay of C’s dispatch has a maximum nodal power-balance residual of 1.76×10^{-15} p.u.

All six price-perturbation checks at 18:00 pass, with a maximum central-difference relative discrepancy of 0.03524%. Together, these residual and independent sensitivity checks support the signs, units, and numerical accuracy of the AC prices used in the joint comparison.

4 Conclusion

The 24-hour joint validation shows close price agreement between the LinDistFlow–Softplus model and full nonlinear AC OPF: the maximum hourly normalized nodal-price error is 1.672%, with no appreciable additional C–B discrepancy. Independent AC checks confirm dispatch feasibility throughout the 24-hour test. These results jointly support the price accuracy and operational feasibility of Model C.